5310 TOTAL ORGANIC CARBON (TOC)

5310 A. Introduction

1. General Discussion

The organic carbon in water and wastewater is composed of a variety of organic compounds in various oxidation states. Some of these carbon compounds can be oxidized further by biological or chemical processes, and the biochemical oxygen demand (BOD) and chemical oxygen demand (COD) may be used to characterize these fractions. The presence of organic carbon that does not respond to either the BOD or COD test makes them unsuitable for the measurement of total organic carbon. Total organic carbon (TOC) is a more convenient and direct expression of total organic content than either BOD or COD, but does not provide the same kind of information. If a repeatable empirical relationship is established between TOC and BOD or COD, then TOC can be used to estimate the accompanying BOD or COD. This relationship must be established independently for each set of matrix conditions, such as various points in a treatment process. Unlike BOD or COD, TOC is independent of the oxidation state of the organic matter and does not measure other organically bound elements, such as nitrogen and hydrogen, and inorganic that can contribute to the oxygen demand measured by BOD and COD. TOC measurement does not replace BOD and COD testing.

To determine the quantity of organically bound carbon, the organic molecules must be broken down to single carbon units and converted to a single molecular form that can be measured quantitatively. TOC methods utilize heat and oxygen, ultraviolet irradiation, chemical oxidants, or combinations of these oxidants to convert organic carbon to carbon dioxide (CO_2). The CO_2 may be measured directly by a nondispersive infrared analyzer, it may be reduced to methane and measured with a flame ionization detector, or CO_2 may be titrated chemically.

2.Fractions of Total Carbon

The methods and instruments used in measuring TOC analyze fractions of total carbon (TC) and measure TOC by two or more determinations. These fractions of total carbon are defined as: inorganic carbon (IC) - the carbonate, bicarbonate, and dissolved CO_2; total organic carbon (TOC) - all carbon atoms covalently bonded in organic molecules; dissolved organic carbon (DOC) - the fraction of TOC that passes through a 0.45 - µm - pore-diam filter; particulate organic carbon (POC) - also referred to as nondissolved organic carbon, the fraction of TOC retained by a 0.45-µm filter; volatile organic carbon (VOC) - also referred to as purgeable organic carbon, the fraction of TOC removed from an aqueous solution by gas stripping under specified conditions; and nonpurgeable organic carbon (NPOC) - the fraction of TOC not removed by gas strlpping.

In most water samples, the IC fraction is many times greater than the TOC fraction. Eliminating or compensating for IC interferences requires multiple determinations to measure true TOC. IC interference can be eliminated by acidifying samples to pH 2 or less to convert IC species to CO_2. Subsequently, purging the sample with a purified gas removes the CO_2 by volatilization. Sample purging also removes POC so that the organic carbon measurement made after eliminating IC interferences is actually a NPOC determination; determine VOC to measure true TOC. In many surface and ground waters the VOC contribution to TOC is negligible. Therefore, in practice, the NPOC determination is substituted for TOC.

Alternatively, IC interference may be compensated for by separately measuring total carbon (TC) and inorganic carbon. The difference between TC and IC is TOC.

The purgeable fraction of TOC is a function of the specific conditions and equipment employed. Sample temperature and salinity, gas-flow rate, type of gas diffuser, purging-vessel dimensions, volume purged, and purging time affect the division of TOC into purgeable and nonpurgeable fractions. When separately measuring VOC and NPOC on the same sample, use identical conditions for purging during the VOC measurement as in purging to prepare the NPOC portion for analysis. Consider the conditions of purging when comparing VOC or NPOC data from different laboratories or different instruments.

3. Selection of Method

The combustion-infrared method (B) is suitable for samples with TOC = 1 mg/L. For lower concentrations, use the persulfate-ultraviolet oxidation method (C) or the wet-oxidation method (D). See 5310B.1, C.1, and D.1.

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5310 B. Combustion-Infrared Method

1. General Discussion

The combustion-infrared method has been used for a wide variety of samples, but its accuracy is dependent on particle size reduction because it uses small-orifice syringes.

a.Principle: The sample is homogenized and diluted as necessary and a microportion is injected into a heated reaction chamber packed with an oxidative catalyst such as cobalt oxide. The water is vaporized and the organic carbon is oxidized to CO_2 and H_2O .The CO_2 from oxidation of organic and Inorganic carbon is transported in the carrier-gas streams and is measured by means of a nondispersive infrared analyzer.

Because total carbon is measured, IC must be measured separately and TOC obtained by difference.

Measure IC by injecting the sample into a separate reaction chamber packed with phosphoric acid-coated quartz beads. Under the acidic conditions, all IC is converted to CO_2 which is measured. Under these conditions organic carbon is not oxidized and only IC is measured.

Alternatively, convert inorganic carbonates to CO_2 with acid and remove the CO_2 by purging before sample injection. The sample contains only the NPOC fraction of total carbon; a VOC determination also is necessary to measure true TOC.

b.Interference: Removal of carbonate and bicarbonate by acidification and purging with purified gas results in the loss of volatile organic substances. The volatiles also can be lost during sample blending, particularly if the temperature is allowed to rise. Another important loss can occur if large carbon-containing particles fail to enter the needle used for injection. Filtration, although necessary to eliminate particulate organic matter when only DOC is to be determined, can result in loss or gain of DOC, depending on the physical properties of the carbon-containing compounds and the adsorption of carbonaceous material on the filter, or its desorption from it. Check filters for their contribution to DOC by analyzing a filtered blank. Note that any contact with organic material may contaminate a sample. Avoid contaminated glassware, plastic containers, and rubber tubing. Analyze treatment, system, and reagent blanks.

c.Minimum detectable concentration: 1 mg carbon/L. This can be achieved with most combustion-infrared analyzers although instrument performance varies. The minimum detectable concentration may be reduced by concentrating the sample, or by increasing the portion taken for analysis.

d.Sampling and storage: Collect and store samples in amber glass bottles with TFE-lined cap. Before use, wash bottles with acid, seal with aluminum foil, and bake at 400 °C for at least 1 h. Wash TFE septa with detergent, rinse repeatedly with organic-free water, wrap in aluminum foil, and bake at 100 °C for 1 h. Preferably use thick silicone rubber-backed TFE septa with open ring caps to produce a positive seal. Because the detection limit is relatively high, less rigorous cleaning may be acceptable; use bottle blanks with each set of samples. Use a Kemmerer or similar type sampler for collecting samples from a depth exceeding 2 m. Preserve samples that cannot be examined immediately by holding at 4 °C with minimal exposure to light and atmosphere. Acidification with phosphoric or sulfuric acid to a pH = 2 at the time of collection is especially desirable for unstable samples, and may be used on all samples; acid preservation, however, requires that inorganic carbon subsequently is purged before analysis.

2. Apparatus

a.Total organic carbon analyzer. (*)
b.Syringes: 0 to 50- µL., 0 to 200-µL, 0 to 500-µL, and 0 to 1-mL capacity.(**)
c.Sample blender or homogenizer.
d. Magnetic stirrer and TFE-coated stirring bars.
e. Filtering apparatus and 0.45-µm-pore-diam filters.

(*) Beckman Instruments Model No. 915B or equivalent

(**) Hamilton No. 705 N : CR-700-20 or CR-700-200 with needle point style No. 3, or equivalent . Spring – loaded syringe may improve reproducibility .

3. Reagents

a.Reagent water: Prepare blanks and standard solutions from carbon-free water; preferably use carbon-filtered, redistilled water.

b. Phosphoric acid, H_3PO_4, conc. Alternatively use sulfuric acid, H_2SO_4 but not hydrochloric acid.

c.Organic carbon stock solution: Dissolve 2.1254 g anhydrous potassium biphthalate, $C_8H_5KO_4$ in carbon-free water and dilute to 1000 mL; 1.00 mL = 1.00 mg carbon. Alternatively, use any other organic-carbon-containing compound of adequate purity, stability, and water solubility. Preserve by acidifying with H_3PO_4 or H_2SO_4 to pH = 2.

d.Inorganic carbon stock solution: Dissolve 4.4122 g anhydrous sodium carbonate, Na_2CO3, in water, add 3.497 g anhydrous sodium bicarbonate,$NaHCO_3$ and dilute to 1000 mL; 1.00 mL = 1.00 mg carbon. Alternatively, use any other inorganic carbonate compound of adequate purity, stability, and water solubility. Keep tightly stopper. Do not acidify.

e.Carrier gas: Purified oxygen or air, CO_2-free and containing less than 1 ppm hydrocarbon (as methane).

f. Purging gas: Any gas free of CO_2 and hydrocarbons.

4. Procedure

a. Instrument operation: Follow manufacturer's instructions for analyzer assembly, testing, calibration, and operation. Adjust to optimum combustion temperature (typically 900 °C) before using instrument; monitor temperature to insure stability.

b.Sample treatment.- If a sample contains gross solids or insoluble matter. homogenize until satisfactory replication is obtained. Analyze a homogenizing blank consisting of reagent water carried through the homogenizing treatment.

If inorganic carbon must be removed before analysis. transfer a representative portion of 10 to 15 mL to a 30-mL beaker, add conc H_3PO_4 to reduce pH to 2 or less, and purge with gas for 10 min. Do not use plastic tubing. Inorganic carbon also may be removed by stirring the acidified sample in a beaker while directing a stream of purified gas into the beaker. Because volatile organic carbon will be lost during purging of the acidified solution, report organic carbon as total nonpurgeable organic carbon.

If the available instrument provides for a separate determination of inorganic carbon (carbonate, bicarbonate, free CO_2) and total carbon, omit decarbonation and proceed according to the manufacturer's directions to determine TOC by difference between TC and IC.

If dissolved organic carbon is to be determined, filter sample through 0.45-µm-pore-diam filter with vacuum; analyze a filtering blank.

c.Sample injection: Withdraw a portion of prepared sample using a syringe fitted with a blunt-tipped needle. Select sample volume according to manufacturer's direction. Stir samples containing particulates with a magnetic stirrer. Select needle size consistent with sample particulate size. Inject samples and standards to analyzer according to manufacturer's directions and record response. Repeat injection until consecutive peaks are obtained that are reproducible to within 10%.

d.Preparation of standard curve: Prepare standard organic and inorganic carbon series by diluting stock solutions to cover the expected range in samples. Inject and record peak height or area of these standards and a dilution water blank. Plot carbon concentration in milligrams per liter against corrected peak height or area on rectangular coordinate paper. This is unnecessary for instruments provided with a digital readout of concentration. If desirable, prepare a standard curve having concentrations of 1 to 10 mg/L by making appropriate dilutions of the standards.

With most TOC analyzers, it is not possible to determine separate blanks for reagent water, reagents, and the entire system. In addition, some TOC analyzers produce a variable and erratic blank that cannot be corrected reliably. In many laboratories, reagent water is the major contributor to the blank value. Correcting only the peak heights of standards (which contain reagent water + reagents + system blank) creates a positive error, while also correcting samples (which contain only reagents and system blank contributions) for the reagent water blank creates a negative error. Minimize errors by using reagent water and reagents low in carbon.

Inject samples and procedural blanks (consisting of reagent water taken through any pre-analysis steps - values are typically higher than those for reagent water) and determine sample organic carbon concentrations by comparing corrected peak heights to the calibration curve.

5.Calculations

a.When reagent water is a major portion of the total blank:

1)Calculate corrected peak height of standards by subtracting the reagent-water blank peak height from the standard peak heights.

2) Prepare a standard curve of corrected peak height vs. TOC concentration.

3)Subtract the procedural blank from each sample peak height and compare to the standard curve to determine carbon content.

NOTE:There will be a positive error if the TOC of the reagent water is significant in comparison to the TOC of the sample. Make a special effort to obtain carbon-free reagent water.

4)Apply the appropriate dilution factor when necessary.

5)Subtract the inorganic carbon from the total carbon when TOC is determined by difference.

b. When reagent water is a minor portion of the total blank:

1)Calculate corrected peak height of standards and samples by subtracting the reagent-water blank peak height from the standard and sample peak heights.

2)Prepare a standard curve of corrected peak height vs. TOC concentration.

3)Subtract the procedural blank from each sample peak height and compare to the standard curve to determine the carbon content. Values will have a negative error equal to the blank contribution from the reagent water.

4)Apply the appropriate dilution factor when necessary.

5)Subtract the inorganic carbon from the total carbon when TOC is determined by difference.

NOTE: If the TOC analyzer design permits isolation of each of the contributions to the total blank, apply appropriate blank corrections to peak heights of standards (reagent blank, water blank, system blank) and sample (reagent blank and system blank).

6. Precision

The difficulty of sampling particulate matter on unfiltered samples limits the precision of the method to approximately 5 to 10%. On clear samples or on those that have been filtered before analysis, precision approaches 1 to 2% or ± 1 to 2 mg carbon / L, whichever is greater.

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5310 C. Persulfate-Ultraviolet Oxidation Method

1. General Discussion

Many instruments utilizing persulfate oxidation of organic carbon are available. They depend either on heat or ultraviolet irradiation activation of the reagents. The per -sulfate-ultraviolet oxidation method is a rapid, precise method for the measurement of trace levels of organic carbon in water and is of particular interest to the electronic, pharmaceutical, and steam-power generation industries where even trace concentrations of organic compounds may degrade ion-exchange capacity, serve as a nutrient source for biological growth, or be detrimental to the process for which the water is being utilized.

a.Principle: Organic carbon is oxidized to carbon dioxide, CO_2, by persulfate in the presence of ultraviolet light. The CO_2 produced may be measured directly by a nondispersive infrared analyzer, be reduced to methane and measured by a flame ionization detector, or be chemically nitrated.

Instruments are available that utilize an ultraviolet lamp submerged in a continuously gas-purged reactor that is filled with a constant-feed per sulfate solution. The samples are introduced serially into the reactor by an auto sampler or they are injected manually. The CO_2 produced is spared continuously from the solution and is carried in the gas stream to an infrared analyzer that is specifically tuned to the absorptive wavelength of CO_2 .The instrument's microprocessor calculates the area of the peaks produced by the analyzer, compares them to the peak area of the calibration standard stored in its memory, and prints out a calibrated organic carbon value in milligrams per liter.

b.Interferences: See Section 5310B.1. Excessive acidification of sample, producing a reduction in pH of the persulfate solution to 1 or less, can result in sluggish and incomplete oxidation of organic carbon.

The intensity of the ultraviolet light reaching the sample matrix may be reduced by highly turbid samples or with aging of the ultraviolet source, resulting in sluggish oxidation . Large organic particles or very large or complex organic molecules such as tannins, lignins, and humic acid may be oxidized slowly because persulfate oxidation is rate-limited. Because the efficiency of conversion of organic carbon to CO_2 may be affected by many factors, check efficiency of oxidation with selected model compounds representative of the sample matrix.

Per sulfate oxidation of organic molecules is slowed in samples containing significant concentrations of chloride by the preferential oxidation of chloride; at a concentration of 0.1% chloride, oxidation of organic matter may be inhibited completely. To remove this interference add mercuric nitrate to the per sulfate solution.

With any organic carbon measurement, contamination during sample handling and treatment is a likely source of interference. This is especially true of trace analysis. Take extreme care in sampling, handling, and analysis of samples below 1 mg TOC/ L.

c.Minimum detectable concentration: Concentration of 0.05 mg organic carbon/L can be measured if scrupulous attention is given to minimizing sample contamination and method background. More typical blank concentrations are 0.2 to 0.3 mg/L; a reporting level of 0.5 mg/L or greater is usual. Use the combustion-infrared method (B) for high concentrations of TOC.

d.Sampling and storage: See Section 5310B. 1d.

2. Apparatus

a.Total organic carbon analyzer. (*)

b.Syringes: 0 to 50-µL, 0 to 250 -µL, and 0 to 1-mL capacity, fitted with a blunt-tipped needle. .

(*) Xertex – Dohrmann DC-80 or equivalent

3. Reagents

See Section 5310B.3.

4. Procedure

a. Instrument operation: Follow manufacturer's instructions for assembly, testing, calibration, and operation.

b.Sample preparation: If a sample contains gross particulates or insoluble matter, homogenize until a representative portion can be withdrawn through the syringe needle or autosampler tubing.

If dissolved organic carbon is to be determined, filter sample and a reagent water blank through 0.7-µm glass fiber filter with vacuum. Pretreat filter submerging overnight in a 1: 1 solution of HNO_3 and reagent water and collect in an acid-washed, baked flask. Use a clean filter and flask for each sample.

To determine NPOC, transfer 15 to 30 mL sample to a flask or test tube and acidify to a pH of 2. Purge according to manufacturer's recommendations.

c. Sample injection: See Section 5310B.4c.

d. Standard curve preparation: Prepare an organic carbon standard series over the range of organic carbon concentrations in the samples. Inject standards and blanks and record analyzer's response. Determine peak area for each standard and blank. Determinations based on peak height may be inadequate because of differences in the rate of oxidation of standards and samples. Correct peak area of standards by subtracting reagent water blank and plot organic carbon concentration in miligrams per liter against corrected peak area. For instruments providing a digital readout of concentration, this is not necessary.

Inject samples, treatment blanks, if applicable, and instrument blank. Subtract appropriate blank's peak area from each sample's peak area and determine organic carbon from the standard curve.

5.Calculation

See Section 5310B.5, but use peak area rather than peak height.

6.Precision and Bias

See Table 5310:I .

TABLE 5310 : I . Precision and Bias for Total Organic Carbon (TOC) by Persulfate –Ultraviolet Oxidation

Characteristics of Analysis	Spring Water	Spring Water + 0.15 mg/L KHP	Tap Water	Tap Water + 10 mg / L KHP	Municipal Wastewater Effluent
Concentration Determined , mg / L : Replicate 1	0.402	0.559	2.47	11.70	5.88
Replicate 2	0.336	0.491	2.49	11.53	5.31
Replicate 3	0.340	0.505	2.47	11.70	5.21
Replicate 4	0.341	0.523	2.47	11.64	5.17
Replicate 5	0.355	0.542	2.46	11.55	5.10
Replicate 6	0.366	0.546	2.46	11.68	5.33
Replicate 7	0.361	0.548	2.42	11.55	5.35
Mean , mg/L	0.35	0.53	2.46	11.53	5.32

Standard Deviation : mg/L	0.02	0.03	0.02	0.21	0.23
%	6	6	1	2	4
Actual value , mg /L	-	0.50	-	12.46	-
Recovery , %	-	106	-	93	-
Error , %	-	6	-	7	-

Note : KHP = Potassium Acid Phthalate

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5310 D. Wet-Oxidation Method

1. General Discussion

The wet-oxidation method is suitable for the analyses of water, water-suspended sediment mixtures, brines, and wastewaters containing at least 0.1 mg nonpurgeable organic carbon (NPOC)/ L. The method is not suitable for the determination of volatile organic constituents.

a. Principle: The sample is acidified, purged to remove inorganic carbon, and oxidized with persulfate in an autoclave at temperatures from 116 to 130 °C. The resultant carbon dioxide (CO_2) is measured by nondispersive infrared spectrometry.

b.Interferences: See Section 5310B.1 and C.1.

c.Minimum detectable concentrations: High concentrations of reducing agents may interfere. Concentrations of 0.10 mg organic carbon/L can be measured if scrupulous attention is given to minimizing sample contamination and method background. Use the combustion-infrared method (B) for high concentrations of TOC.

d.Sampling and storage: See Section 5310B. 1d.

2. Apparatus

a.Ampules, precombusted, 10-mL, glass. (*)
b.Ampule purging and sealing unit. (*)
c.Autoclave. (*)
d.Carbon anlyzer. (*)
e.Homogenizer. (**)
(*) Oceanographics International Corp. or equivalent .
(**) Willems Polytron PT – 105 T , Brinkman Inst. , or equivalent .

3. Reagents

In addition to the reagents specified in Section 5310B. *3a, c, e,* and *f,* the following reagents are required:

a.Phosphoric acid solution, H_3PO_4 1.2*N*: Add 83 mL H_3PO_4 (85%) to water and dilute to 1 L with water. Store in a tightly stoppered glass bottle.

b.Potassium persulfate, reagent-grade, granular. Avoid using finely divided forms.

4. Procedure

a.Instrument operation: Follow manufacturer's instructions for assembly, testing calibration, and operation. Add 0.2 g potassium persulfate using a dipper calibrated to deliver 0.2 g and 0.5 mL 1.2N H_3PO_4 solution to precombusted ampoules.

To analyze for dissolved organic carbon, pass sample and reagent water blank under vacuum through a separate 0.45-µm glass fiber filter that was pretreated by submerging overnight in a 1:1 solution of $HN0_3$ and reagent water and collect in acid-washed, baked flasks. Use clean filter and flask for each sample.

Homogenize sample to produce a uniform suspension. Rinse homogenize with reagent water after each use. Pipette water sample (10.0 mL maximum) into an ampule. Adjust smaller volumes to 10 mL with reagent water. Prepare one reagent blank (10 mL reagent water plus acid and oxidant) for every 15 to 20 water samples. Prepare standards covering the range of 0.1 to 40 mg carbon/L by diluting the carbon standard solution. Immediately place filled ampoules on purging and sealing unit and purge them at rate of 60 min/min for 6 min with purified oxygen. Seal samples according to the manufacturer's instructions. Place sealed samples, blanks, and a set of standards in ampoule racks in an autoclave and digest 4 h at temperature between 116 and 130 °C.

Set sensitivity range of carbon analyzer by adjusting the zero and span controls in accordance with the manufacturer's instructions. Break combusted ampoules in the cutter assembly of the carbon analyzer, sweep CO_2 into the infrared cell with nitrogen gas, and record area of each CO_2 peak. CAUTION: *Because combusted ampoules are under positive pressure, handle with care to prevent explosion.*

5.Calculations

Prepare an analytical standard curve by plotting peak area of each standard versus concentration (mg/L) of organic carbon standards. The relationship between peak area and carbon concentration is curvilinear. Define operating curves each day samples are analyzed.

Report nonpurgeable organic carbon concentration (NPOC) as follows: 0.1 mg/L to 0.9 mg/L, one significant figure; 1.0 mg/L and above, two significant figures.

6.Precision and Bias

Multiple determinations of four different concentrations of aqueous potassium acid phthalate samples at 2.00, 5.00, 10.0, and 40.0 mg carbon/L resulted in mean values of 2.2, 5.3, 9.9, and 38 mg/L and standard deviations of 0.13, 0.15, 0.11, and 1.4, respectively.

Precision also may be expressed in terms of percent relative standard deviation as follows:

Number of Replicates	Mean (mg / L)	Relative Standard Deviation %
9	2.2	5.9
10	5.3	2.8
10	9.9	1.1
10	38.0	3.7

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